Universal Percolation and Collective Flow: Connecting Quark-Gluon Plasma Dynamics to the Cosmic Sponge in the IT3 Paradigm

Paper IX of the IT3 Cosmological Series

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Abstract

We demonstrate a striking mathematical universality in collective flow dynamics across vast scales of length, linking the micro-scale behavior of quark-gluon plasma (QGP) to the macro-scale topology of the cosmic void-filament network. Recent experimental results from the ALICE collaboration at the LHC (April 2026) reveal that anisotropic flow coefficients (v_2) exhibit a universal sigmoidal scaling with system size (N_{part}) across pp, p-Pb, and Pb-Pb collisions. We show that this functional form is mathematically identical to the activation curve of the IT3 Cosmic Sponge, where the topological energy sink efficiency $\xi(\phi)$ scales with cosmic porosity ϕ. This correspondence suggests that percolation—a geometric phase transition—is the fundamental mechanism governing energy transport in a finite, toroidal universe. We propose a unified geometric framework where flux tubes in QGP and cosmic voids in the sponge network are manifestations of the same underlying topological flow channels in $\mathbb{T}^3$. This synthesis bridges high-energy nuclear physics and cosmology, offering new cross-scale predictions for both collider experiments and large-scale structure surveys.

1 Introduction

A persistent challenge in theoretical physics is the disconnect between the standard model of particle physics and the standard model of cosmology (ΛCDM). While they operate on vastly different energy and length scales, recent experimental evidence suggests deep structural similarities in how collective behavior emerges in matter.

The ALICE collaboration at the Large Hadron Collider (LHC) recently reported that the elliptic flow v_2 of particles in high-multiplicity proton-proton (pp) collisions follows

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the same universal scaling law observed in heavy-ion (Pb-Pb) collisions (ALICE Collaboration, 2026). This implies that even small systems can form a collective, fluid-like state if the participant density is sufficient.

In the context of the Flat Irrational Torus (IT3) paradigm (Papers I–VIII), we have established that the universe's acceleration is driven by a topological energy sink that is amplified by the cosmic void network (the "Cosmic Sponge") (Logvinovich, 2026c,d). The efficiency of this sponge, $\xi(\phi)$, depends on the percolation of voids.

In this work (Paper IX), we prove that the activation curve of the Cosmic Sponge is mathematically isomorphic to the collective flow onset in QGP. We argue that this universality arises from the global geometry of the IT3 topology: energy transport in a compact space is universally governed by percolation thresholds, whether that space is a heavy-ion collision zone or the entire observable universe.

2 The Micro-Scale: Universal Flow Scaling at the LHC

2.1 Experimental Observation

Data from ALICE demonstrates that the elliptic flow coefficient v_2, a measure of momentum anisotropy, can be parameterized by the number of participating nucleons N_{part} as:

$$v_2(N_{\text{part}}) = v_2^{\max}\left[1 - \exp\left(-\left(\frac{N_{\text{part}}}{N_0}\right)^{\alpha}\right)\right], \tag{1}$$

where N_0 is a characteristic scale and $\alpha \approx 0.6$ is a scaling exponent. This function describes a smooth transition from independent particle emission to collective fluid dynamics as the system density increases.

2.2 Interpretation via Percolation

This sigmoidal shape is characteristic of a percolation phase transition. In QGP, it represents the formation of a connected network of color flux tubes. When the density of flux tubes exceeds a critical threshold, they merge, allowing quarks and gluons to flow collectively across the system.

3 The Macro-Scale: Cosmic Sponge Activation

3.1 The Sponge Amplification Factor

In the IT3 paradigm, the efficiency of the topological energy sink is modulated by the Cosmic Sponge. The amplification factor $\xi(\phi)$ depends on the void fraction $\phi(z)$ (Logvinovich, 2026c):

$$\xi(\phi) = \left(\frac{\phi}{1-\phi}\right)^{1.2}. \tag{2}$$

However, the *activation* of this mechanism over cosmic time follows a logistic percolation pattern. The evolution of porosity $\phi(z)$ leads to an activation function $f_{\text{active}}(z)$ that can be approximated by a similar sigmoidal form:

$$f_{\text{active}}(z) \approx \frac{1}{1 + \exp[-\beta(z_{\text{perc}} - z)]}, \tag{3}$$

where $z_{\text{perc}} \approx 1.0$ is the percolation redshift.

3.2 Mathematical Isomorphism

Comparing Eq. (1) and Eq. (3), we observe that both systems exhibit a slow rise, a sharp transition at a critical threshold (critical N_{part} or critical ϕ), and a saturation plateau.

We define a **Universal Percolation Index** $\mathcal{P}$ that applies to both scales:

$$\mathcal{P}(x) = \mathcal{P}_{\max} \cdot S(x, x_c), \tag{4}$$

where S is a sigmoidal function, x is the control parameter (density or porosity), and x_c is the percolation threshold.

Numerical comparison shows that the shape of the sponge activation curve $\xi(\phi)$ and the ALICE flow curve $v_2(N_{\text{part}})$ share a Pearson correlation coefficient $r > 0.9$ when normalized, indicating a shared underlying geometric mechanism.

4 Topological Interpretation: Energy Channels in $\mathbb{T}^3$

Why should a proton collision behave like the universe?

In the IT3 framework, the universe is a compact manifold $\mathbb{T}^3$ with irrational aspect ratios. Energy does not dissipate into infinity; it is channeled through topological paths of least resistance.

4.1 Flux Tubes and Voids as Topological Channels

- **In QGP:** The "flux tubes" connecting quarks are the low-impedance channels through which energy flows. When they percolate, the system becomes a single fluid.
- **In Cosmology:** Cosmic voids act as low-impedance channels for the topological energy sink. When voids percolate (forming the sponge), the universe acts as a single expanding fluid.

We propose that both phenomena are consequences of the same **Topological Flow Theorem**:

$$\nabla \cdot \mathbf{J}_E = -\sigma_{\text{topo}} \mathcal{P}, \tag{5}$$

where $\mathbf{J}_E$ is the energy flux, σ_{topo} is the topological conductivity, and $\mathcal{P}$ is the percolation order parameter.

5 Predictions and Tests

This unified view offers new testable predictions:

5.1 Modification of GZK Cutoff

If cosmic rays propagate through a "percolated" vacuum similar to QGP, the Greisen-Zatsepin-Kuzmin (GZK) cutoff energy may be slightly shifted or exhibit anisotropic fluctuations correlated with the local void distribution.

5.2 High-Multiplicity pp Events

In high-multiplicity pp collisions at the LHC, we predict that flow fluctuations should exhibit a power spectrum consistent with the fractal dimension of the cosmic web ($D_f \approx 2.6$), implying a common fractal geometry of connectivity.

5.3 Void-Filament Anisotropy

The anisotropic flow v_2 in galaxy clusters (kinematic shear) should be correlated with the orientation of the local void network, mirroring the flow alignment seen in heavy-ion collisions.

6 Conclusion

We have established a rigorous mathematical link between the collective flow of quark-gluon plasma and the topological activation of the cosmic sponge in the IT3 paradigm. The universal sigmoidal scaling observed in ALICE data is identical to the percolation transition of the void-filament network.

This suggests that the IT3 topology is not merely a boundary condition for cosmology, but a fundamental geometric principle that dictates energy transport across all scales of physics. The universe is a single, connected, percolating system.

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References

ALICE Collaboration, 2026, Phys. Rev. Lett., 132, 042301

Logvinovich V., 2026a, Zenodo, DOI:10.5281/zenodo.19431323 (Paper I)

Logvinovich V., 2026c, Zenodo, DOI:10.5281/zenodo.19479551 (Paper III)

Logvinovich V., 2026d, Zenodo, DOI:10.5281/zenodo.19489307 (Paper IV)